Today: - convergence theorem for CTMCs:

- a finite irreducible CTMC satisfies

$$p_t(x,y) \xrightarrow{t\to\infty} \pi(y) \quad \forall x$$

where π unique stat dist.

- frequency of visits: under same conditions,

Θ capital theta

$$\frac{1}{t}\,\Theta_t(x) \xrightarrow{t\to\infty} \pi(x)$$

occupation time

$$\Theta_t(x) = \int_0^t \mathbb{1}(X_s = x)\,ds$$

- hitting probability & hitting time:
formulas & examples.

<u>Convergence:</u>

irreducibility: a CTMC is irreducible if $\forall x, y$
$\exists\ t:\quad p_t(x,y) > 0.$

exercise: a CTMC (X_t) is irred. $\Leftrightarrow$ its discrete-time embedded Markov chain (Y_n) is irreducible.

in continuous time there is no periodicity:

if a CTMC is irreducible then

$\forall t > 0$ $\quad p_t(x,y) > 0.$ $\quad (*)$

(pf. in book)

Convergence theorem: given irred. CTMC, fix $h > 0$ and examine DTMC with trans. mtx.

$p_h(x,y)$

from $(*)$, p_h is irreducible & aperiodic

$p_h(x,x) > 0 \quad \forall x.$

$\forall\, x, y,$

$$p_{nh}(x,y) \xrightarrow{n \to \infty} \underline{\pi(y)}$$

stat. dist. of p_h

$=$

... stat. dist. of CTMC

if $\pi Q = 0$

then $\forall t \quad \pi p_t = \pi$

Occupation time

☆ two ingredients:

- if (X_t) has embedded chain (Y_n) with stat dist ν then (X_t) has stat. dist.

$$\pi(x) = C \, \nu(x) / \lambda_x$$

proportion of all jumps that go to x

$1/\lambda_x$: exp. time spent @ x per visit.

- C: chosen so

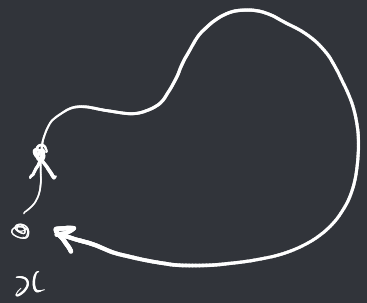
$$\sum_x \pi(x) = 1$$

- from the renewal-reward theorem,

$$\frac{1}{t} \textcircled{H}_t(x) \xrightarrow{t \to \infty} \frac{\left(\frac{1}{\lambda_x}\right)}{E_x[T_x^+]} = \lambda_x / \pi(x)$$

need to connect to $\pi(x)$

want



T_x^+ return time

$= \inf\{t : X_t = x,\ X_s \neq x \text{ some } s < t\}$.

reward / excursion

= time at x

which has $E[\cdot] = 1/\lambda_x$. (recall $\lambda_x = \sum_y q(x,y)$)

we know that in discrete time (use $\tilde{E}$ for (Y_n))

$$\tilde{E}_x[T_x^+] = 1/v(x)$$

and we have (using E for (X_t))

$$E_x[T_x^+] = \sum_y \underbrace{E_x\left[\Theta_{T_x^+}(y)\right]}$$

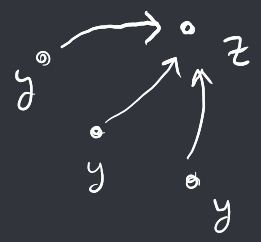
$$= \underbrace{\tilde{E}_x\left[N_{T_x^+}(y)\right]}_{= v(y)\,\tilde{E}_x[T_x^+] = v(y)/v(x)} \cdot 1/\lambda_y$$

(construction of v from excursion)

$$= \sum_y \frac{\nu(y)}{\nu(x)} \cdot \frac{1}{\lambda_y}$$

$$= \sum_y \frac{\pi(y)}{\pi(x)} \cdot \lambda_x$$

$$= \lambda_x / \pi(x)$$



Hitting probability & hitting time

recall for A, B : $A \cap B = \phi$,

$$h(x) := P_x(X_\tau \in A) \qquad \tau = \inf\{t : X_t \in A \cup B\}$$

then

$$\sum_y Q(x,y)\, h(y) = 0 \qquad \forall x \notin A \cup B$$

$$\Leftrightarrow$$

$$\sum_{y \neq x} q(x,y)\, h(y) - \lambda_x h(x) = 0$$

$$\Leftrightarrow$$

$$h(x) = \sum_{y \neq x} \frac{q(x,y)}{\lambda_x} h(y)$$

= trans. prob. of embedded chain.

h has the property that

$$M(t) := h(\underbrace{X_{t \wedge \tau}}_{\min(t,\tau)}) \quad \text{is a martingale.}$$

Stopping at τ means we never need to take

$$(Qh)(x) \quad \text{for } x \in A \cup B$$

when calculating average change of M

(recall for any f,

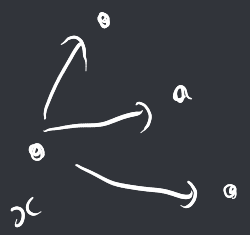
$$f(X_t) - f(X_0) - \int_0^t \underbrace{(Lf)(X_s)}\, ds$$

& we know $(Lh)(x) = 0$ for $x \notin A \cup B$.

<u>Hitting time:</u>

fix $A \subset S$. Define $\tau = \inf\{t : X_t \in A\}$.

let $g(x) := E_x[\tau]$.



in discrete time we had

$$g(x) = 1 + \sum_y p(x,y)\, g(y)$$

$$\forall x \notin A.$$

In continuous time,

$$g(x) = 1/\lambda_x + \sum_y p(x,y)\, g(y)$$

This also has a "matrix" form:

$$\lambda_x g(x) = 1 + \sum_y q(x,y)\, g(y)$$

$$0 = 1 + (Qg)(x)$$

$$\therefore \quad \underbrace{(Qg)(x)}_{= (Lg)(x)} = -1 \qquad \forall\, x \notin A.$$

recall $(Lf)(x) = \sum_{y \neq x} (f(y) - f(x))\, q(x,y)$

$$= \sum_{y \neq x} q(x,y) f(y) - f(x) \lambda_x$$

$$= (Qf)(x).$$

<u>examples</u>

$\rightarrow$ find $P_x(\text{hit } N \text{ before } 0)$ in br. process

with $p_0 = 1/2$, $p_2 = 1/2$.

(recall $z \to z+1$ at rate $z/2$
$z \to z-1$ " " $z/2$)

$$z-1 \quad \overset{\curvearrowleft}{\quad z \quad} \overset{\curvearrowright}{\quad z+1}$$

from rates $(Qh)(x) = 0$, $0 < x < N$

$$0 = \sum_x q(x,y)h(y) = -\lambda_x h(x) + \frac{x}{2} h(x-1) + \frac{x}{2} h(x+1)$$

$$= -xh(x) + \frac{x}{2}h(x-1) + \frac{x}{2}h(x+1)$$

$$\therefore \quad h(x) = \frac{1}{2}\left(h(x-1) + h(x+1)\right)$$

(same as g.r. with $p = \frac{1}{2}$ because embedded chain has $p(x, x+1) = \frac{q(x, x+1)}{\lambda x}$

$$= \frac{x/2}{x} = \frac{1}{2}$$

similarly $p(x, x-1) = \frac{1}{2}$.

so $P_x(\text{hit } N \text{ before } 0) = x/N$ as in g.r. with $p = 1/2$.

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what if $p_2 = p$, $p_0 = 1-p$?

then

$$Z \to Z+1 \quad \text{at rate} \quad pZ$$
$$Z \to Z-1 \quad \text{"} \quad \text{"} \quad (1-p)Z$$

embedded chain has $p(x, x+1) = p$ $p(x, x-1) = 1-p$.

$\therefore$ same hitting probab. as g.r. with "up" probab. $= p$.

what about expected hitting time?

in contrast to hitting probab, can't completely reduce to embedded chain.

if transition rates independent of state, then you still can.

ex: CTMC with $q(x, x+1) = c_1$

$q(x, x-1) = c_2$

has for any A, τ & $g(x) = E_x[\tau]$

$\forall\, x \notin A$, $\quad (2) \quad g(x) = 1/\lambda_x + \sum_y p(x,y)\, g(y)$

$$= \underbrace{1/(c_1 + c_2)}$$

when this is a constant can scale the equation:

if $\tilde{g}$ satisfies $\tilde{g}(x)=0, \quad x \in A$

and

$$\tilde{g}(x) = 1 + \sum_y p(x,y)\tilde{g}(y)$$

then $\dfrac{\tilde{g}}{c_1+c_2}$ satisfies (2).

so for example if $c_1 = c_2 = \frac{1}{2}$; random walk in continuous time and

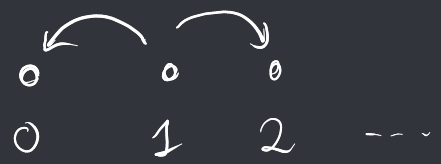
$E_x[\text{time to hit } \{0, N\}]$

same as in discrete time.
(outgoing total rate $= 1$ @ each state).

In some cases we can still appeal to the embedded chain:

<u>ex</u>: show $E_1[\tau_0^+] = \infty$ for br. proc. with $p_0 = p_{1/2} = \frac{1}{2}$.

wrt embedded chain

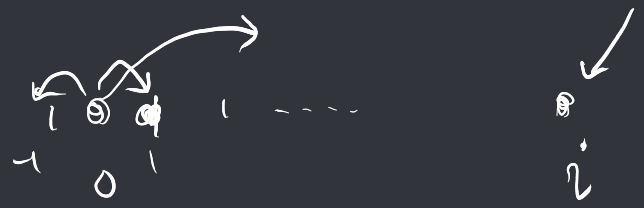


(3) $$E_1[\tau_0^+] = \sum_i \underbrace{\tilde{E}_1[N_{\tau_0^+}(i)]}_{\text{need to compute these.}} \cdot \frac{1}{\lambda_i}$$

<u>fact</u>: (mostly follows from previous homework)

$\forall i, \quad \tilde{E}_1\left[N_{\tau_0^+}(i)\right] = 2.$

why?



exp. ctd # visits to i on excursion from 0 $= 1$.

from 0, jump left:

- no visits to i. (pr $= 1/2$)
  jump right.
- N visits to i (pr $= 1/2$)

exp. ctd # visits $= N/2$ $\therefore$ $N = 2$.

$\therefore \quad (3) = \sum_i 2 \cdot \frac{1}{i} = \infty.$

<u>Notice:</u>

without changing embedded chain if we took instead

$$z \to z+1 \quad \text{at rate } \lambda z/2$$
$$z \to z-1 \quad \text{at rate } \lambda z/2$$

then $(3) = \sum_i 2/\lambda_i$

so taking e.g. $\lambda_i = i^2$,

expected return time is now finite.

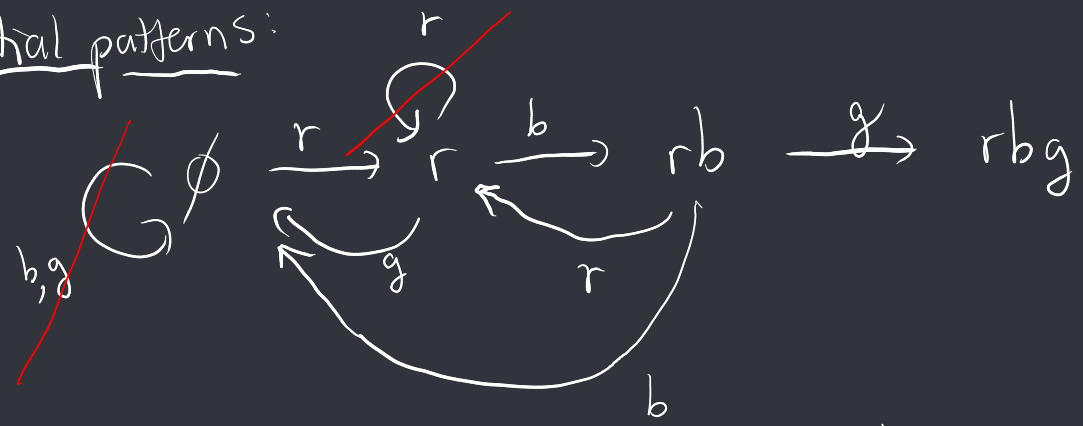
<u>example:</u> pattern time

three types of events occur:

- blue flash at rate λ_b
- red flash " " λ_r
- green flash " " λ_g

find $E[\text{time until consecutive pattern rbg occurs}]$

<u>partial patterns:</u>



$g(rbg) = 0$

$g(x) := E_x[\tau_{rbg}]$

$$g(\phi) = 1/\lambda_r + g(r)$$

$$p(\phi, r) = 1.$$

(exercise: check

$$g(\phi) = \frac{1}{\lambda_r + \lambda_b + \lambda_g} + \frac{\lambda_b + \lambda_g}{\lambda_r + \lambda_b + \lambda_g} g(\phi)$$

$$+ \frac{\lambda_r}{\lambda_r + \lambda_b + \lambda_g} g(r)$$

simplifies to the same formula)

$$g(r) = \frac{1}{\lambda_b + \lambda_g} + g(\phi) \cdot \frac{\lambda_g}{\lambda_b + \lambda_g}$$

$$+ g(rb) \cdot \frac{\lambda_b}{\lambda_b + \lambda_g}$$

$$g(rb) = \frac{1}{\lambda_r + \lambda_b + \lambda_g} + g(r) \cdot \frac{\lambda_r}{\lambda_r + \lambda_b + \lambda_g}$$

$$+ g(\phi) \cdot \frac{\lambda_b}{\lambda_r + \lambda_b + \lambda_g}.$$